

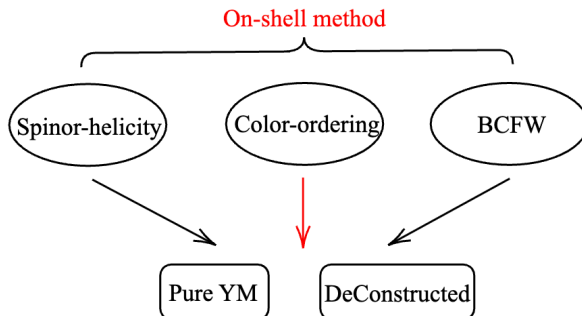
On-Shell Methods for Tree-Level Amplitudes in (De)Constructed Gauge Theory

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I will mainly focus on the color-ordering in this report.

1 Motivation

2 Preliminary

3 Model and Computation

4 Summary and Outlook

Why We Study Scattering Amplitudes?

1. Bridge between theory and experiment

- Core prediction targets for high-energy collider experiments such as the LHC , especially for high multiplicity amplitudes.
- Any new theory (SUSY, GUTs, extra dimensions) must predict observable cross sections

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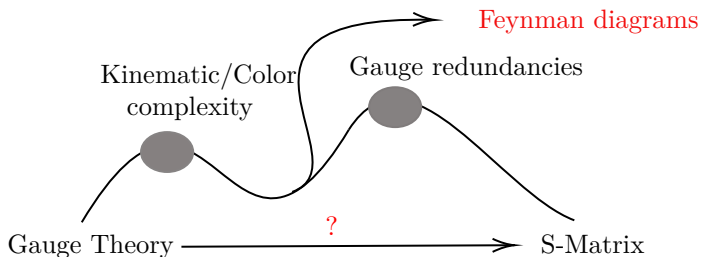
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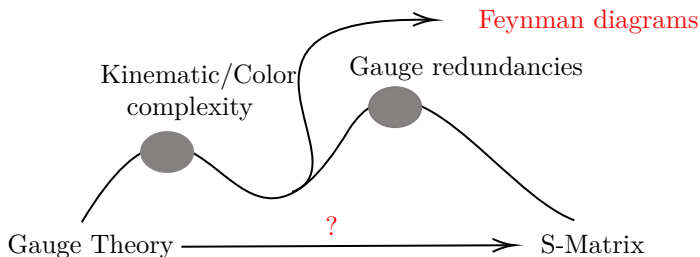
2. Reveal deep structures of quantum field theory

- Amplitudes exhibit hidden symmetries (e.g., dual conformal, Yangian) not visible in the Lagrangian
- These symmetries suggest deeper theoretical frameworks, such as amplituhedra or holographic principle.

Challenges we face before



Challenges we face before

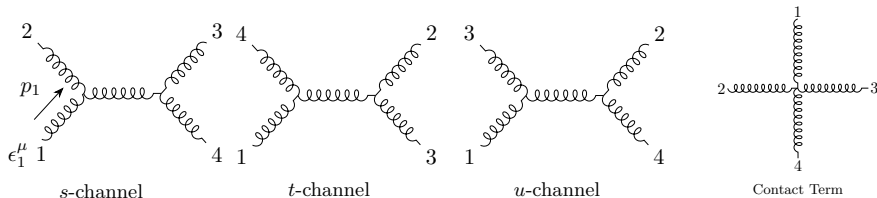


n pt. amplitudes	4	5	6	7	8	9	10
# of diagrams	4	25	220	2485	34300	559405	10525900

The number of Feynman diagrams grow quite rapidly!

Conventional Computation

Usually, when we compute the gluon amplitudes by using Feynman diagram, we will obtain something like



$$\begin{aligned}
 \mathcal{M}_s(p_1 p_2 \rightarrow p_3 p_4) = & -\frac{g_s^2}{s} f^{abe} f^{cde} \\
 & \times \left\{ -4 \epsilon_1 \cdot \epsilon_3^* \epsilon_2 \cdot p_1 \epsilon_4^* \cdot p_3 + 2 \epsilon_1 \cdot \epsilon_2 \epsilon_3^* \cdot p_1 \epsilon_4^* \cdot p_3 \right. \\
 & \left. - 2 \epsilon_1 \cdot p_4 \epsilon_2 \cdot p_1 \epsilon_3^* \cdot \epsilon_4^* + \epsilon_1 \cdot \epsilon_2 p_4 \cdot p_1 \epsilon_3^* \cdot \epsilon_4^* + \text{remaining 10 terms} \right\}
 \end{aligned}$$

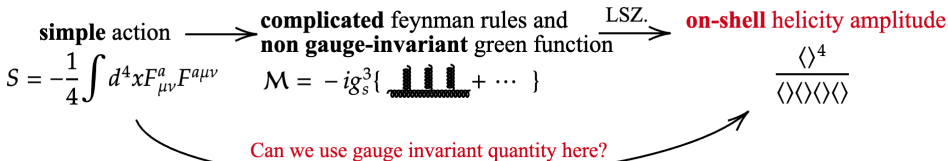
If you consider 5point case, it will become worse:

★ We have 25 diagrams and nearly 10000 terms!



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Gauge Theory $\xrightarrow{\text{On-shell method}}$ Helicity Amplitude

Parke-Taylor Formula (MHV amplitudes):

$$\begin{aligned} A_5[1^+2^+3^+4^+5^+] &= 0 & (+, - : \text{helicity;} \\ A_5[1^-2^+3^+4^+5^+] &= 0 & 1, 2, \dots, n : \text{particle labels}) \end{aligned}$$

$$A_5[1^-2^-3^+4^+5^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle} \quad (\text{first non-trivial one})$$

$$(p_{i\mu} \sigma^\mu)_{\alpha\dot{\alpha}} = (|p_i\rangle)_{\alpha} ([p_i|)_{\dot{\alpha}}, \quad |p_i\rangle = |i\rangle$$

$$\sigma^\mu = (1, \vec{\sigma}), \quad |i\rangle, [i] : \text{Weyl Spinors}$$

$$\langle ij \rangle = \langle i |^{\alpha} | j \rangle_{\alpha} = \varepsilon^{\alpha\beta} |i\rangle_{\beta} |j\rangle_{\alpha}, \quad \varepsilon^{\alpha\beta} : \text{antisymmetric tensor}$$

Modern Amplitude Method

The answer is On-shell method.

Gauge Theory $\xrightarrow{\text{On-shell method}}$ S-Matrix

$$A_5[1^+2^+3^+4^+5^+] = 0$$

$$A_5[1^-2^+3^+4^+5^+] = 0$$

$$A_5[1^-2^-3^+4^+5^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle}$$

$$A_5[1^-2^-3^-4^-5^-] = 0$$

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$$M_5 = \underbrace{A_5[12345]}_{\text{Color-ordered Amplitudes}} \text{Tr}[T^{a_1} T^{a_2} \dots T^{a_5}] + \text{permutations}$$

Color-ordered Amplitudes

Color-ordering for Yang-Mills

Yang-Mills Lagrangian:

$$\mathcal{L} = -\frac{1}{2} \text{Tr}(F_{\mu\nu} F^{\mu\nu})$$

Color factors for s, t, u channels:

$$c_s = \tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4}, \quad c_t = \tilde{f}^{a_4 a_1 b} \tilde{f}^{b a_2 a_3}, \quad c_u = \tilde{f}^{a_1 a_3 b} \tilde{f}^{b a_2 a_4}$$

Using $i\tilde{f}^{abc} = \text{Tr}([T^a, T^b]T^c)$, s -channel becomes:

$$\begin{aligned} \tilde{f}^{a_1 a_2 b} \tilde{f}^{b a_3 a_4} &= \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4}) - \text{Tr}(T^{a_2} T^{a_1} T^{a_3} T^{a_4}) \\ &\quad - \text{Tr}(T^{a_1} T^{a_2} T^{a_4} T^{a_3}) + \text{Tr}(T^{a_2} T^{a_1} T^{a_4} T^{a_3}). \end{aligned}$$

Thus 4-gluon tree amplitude can be rewritten like:

$$\mathcal{A}_4^{\text{tree}} = g^2 \sum_{\sigma \in S_3} A_4[1\sigma(234)] \text{Tr}(T^{a_1} T^{a_{\sigma(2)}} T^{a_{\sigma(3)}} T^{a_{\sigma(4)}})$$

here the kinematic factor $A_4[1234]$, $A_4[1243]$, etc. are called **color-ordered amplitudes**.

General n -point tree-level:

$$\mathcal{A}_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{n-1}} A_n[1, \sigma(2, \dots, n)] \text{Tr}(T^{a_1} T^{a_{\sigma(2)}} \dots T^{a_{\sigma(n)}})$$

Some Properties

- Cyclic: $A_n[12 \cdots n] = A_n[2 \cdots n1]$
- Reflection: $A_n[12 \cdots n] = (-1)^n A_n[n \cdots 21]$
- U(1) decoupling identity:

$$A_n[123 \cdots n] + A_n[213 \cdots n] + A_n[231 \cdots n] + \cdots + A_n[23 \cdots 1n] = 0$$

- Kleiss-Kuijf (KK) relations:

$$A_n[1, \{\alpha\}, n, \{\beta\}] = (-1)^{|\beta|} \sum_{\sigma \in \text{OP}(\{\alpha\}, \{\beta^T\})} A_n[1, \sigma, n]$$

e.g. $A_5[12534] = A_5[12435] + A_5[14235] + A_5[14325]$

- BCJ relations:

$$s_{12}A_5[21345] - s_{23}A_5[13245] - (s_{23} + s_{24})A_5[13425] = 0$$

- Cyclicity: The trace $\text{Tr}(T^{a_{\sigma(1)}} T^{a_{\sigma(2)}} \dots T^{a_{\sigma(n)}})$ itself is cyclic, so in order to make the full amplitude invariant, the color-ordered amplitude in front of the trace should also be cyclic.

$$A[\sigma(1)\sigma(2)\dots\sigma(n)]\text{Tr}(T^{a_{\sigma(1)}} T^{a_{\sigma(2)}} \dots T^{a_{\sigma(n)}})$$

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$$A[\sigma(1)\sigma(2)\dots\sigma(n)]\text{Tr}(T^{a_{\sigma(1)}} T^{a_{\sigma(2)}} \dots T^{a_{\sigma(n)}})$$

- U(1) decoupling identity: If one of the gluon are replaced by photon, the amplitude should vanish because there's no interaction between photon and gluon.

$$\mathcal{A}(1^{\text{photon}}, 2, \dots, n) = 0$$

and the corresponding generator are replaced with identity

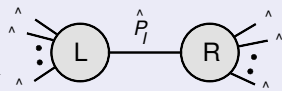
$$\text{Tr}(\mathbf{I}\sigma(2)\sigma(3)\dots\sigma(n)) \quad \text{is invariant if } \sigma \text{ is cyclic}$$

e.g.

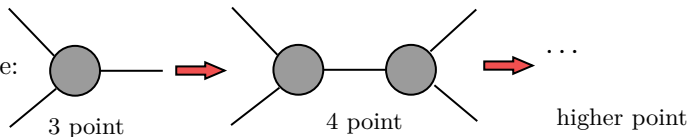
$$\begin{aligned} & \text{Tr}(\mathbf{I}234) = \text{Tr}(\mathbf{I}342) = \text{Tr}(\mathbf{I}423) \\ \Rightarrow & A[1234]\text{Tr}(\mathbf{I}234) + A[1342]\text{Tr}(\mathbf{I}342) + A[1423]\text{Tr}(\mathbf{I}423) \\ & = (A[1234] + A[1342] + A[1423])\text{Tr}(\mathbf{I}234) = 0 \\ \Rightarrow & \mathbf{A[1234] + A[1342] + A[1423] = 0} \end{aligned}$$

The Power of BCFW Recursion Relation

BCFW recursion relation

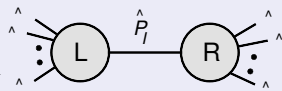
$$A_n = \sum_{\text{diagrams } I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I) = \sum_{\text{diagrams } I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I)$$


More understandable:

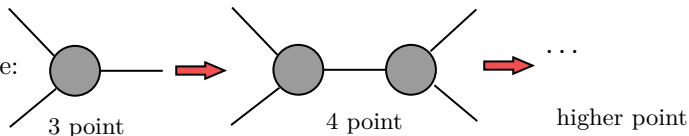


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More understandable:



Specifically, it helps us to prove P. T. formula

$$\begin{array}{ccccccc}
 \text{3pt.} & \longrightarrow & \text{4pt.} & \longrightarrow & \text{5pt.} & \longrightarrow & \dots \quad \text{n pt.} \\
 \frac{\langle 12 \rangle^3}{\langle 23 \rangle \langle 31 \rangle} & & \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} & & \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle} & & \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}
 \end{array}$$

★ This is the power of BCFW recursion relation.

So far: Foundations

- Reviewed the structure of **BCFW recursion relation**
- Applied to:
 - **Pure Yang-Mills** theory
 - Tree-level MHV amplitudes
 - Color-ordered partial amplitudes

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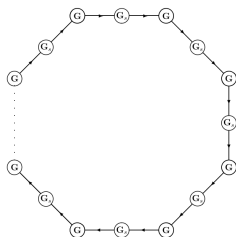
Next: Realistic Models

- Move beyond massless gauge theory
- Consider:
 - **(De)constructed gauge theories**
- Key questions:
 - Can BCFW still apply?
 - What is the color structure in this case?

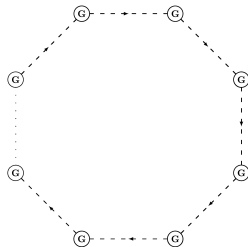
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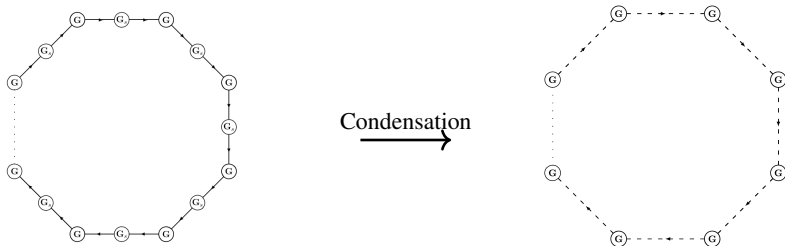
Introduction of (De)Constructed gauge theory



Condensation
 $\longrightarrow$



Introduction of (De)Constructed gauge theory



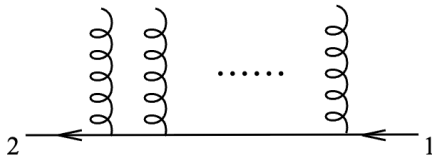
The Lagrangian can be written like

$$\mathcal{L} = - \sum_{i=1}^N \frac{1}{2} \text{Tr}(F_i^2) + \sum_{i=1}^N \text{Tr} \left[(D_\mu \Phi_i)^\dagger (D^\mu \Phi_i) \right],$$

here F_i refers to the i th gauge field strength. The scalar field Φ_i transforms under the **bi-fundamental** representation, and the covariant derivative equals to

$$D_\mu \Phi_i = \partial_\mu \Phi_i - ig_i A_{i\mu} \Phi_i + ig_{i+1} \Phi_i A_{i+1\mu}.$$

SQCD like sector



The color factor in this sector looks like

$$(T^{a_1} T^{a_2} \dots T^{a_n})_{ij}$$

so what we need to notice is just the order of gauge boson.

The amplitudes can be computed like

- $\Phi^\dagger V_1 V_1 \Phi$

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^-] = (-1) \frac{\langle 14 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \quad (\text{Parke -Talyor like Formula})$$

- $\Phi^\dagger V_1 V_1 V_1 \Phi$

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^+ 5^-] = \frac{\langle 15 \rangle^2 \langle 25 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 51 \rangle}$$

- $\Phi^\dagger (n V_1) \Phi$

$$A[1^\Phi 2^{\Phi^\dagger} \cdots (n+2)^-] = (-1)^{n+1} \frac{\langle 1, n+2 \rangle^2 \langle 2, n+2 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n+1, n+2 \rangle \langle n+2, 1 \rangle}$$

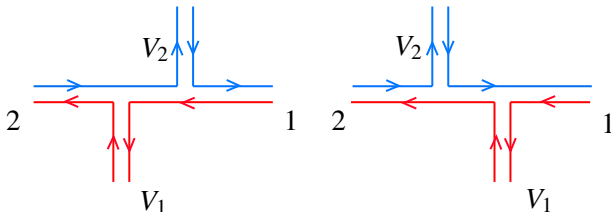
★ Bonus relation:

$$A[1^\Phi 2^{\Phi^\dagger} 3^+ 4^+] = 0 \quad \Rightarrow \quad A[1^\Phi 2^{\Phi^\dagger} 3^+ \cdots n^+] = 0$$

For the amplitude $\Phi(n V_2) \Phi^\dagger$, we can obtain nearly the same expression.

Pure 2-site amplitude

The straightforward way to observe the color structure in this case is double line notation as follows



The color factor here have special form like

$$(T_1^{a_1} T_1^{a_2} \cdots T_1^{a_{n_1}})_{ij} (T_2^{b_1} T_2^{b_2} \cdots T_2^{b_{n_2}})_{\bar{j}\bar{i}}$$

we can notice that the relative order between two gauge group do not affect the color structure, but the order inside the gauge group matters.

So we introduce the **OPP (Order Preserving Permutation)**, for example

$$\begin{array}{lll} (3_1^+, 4_1^+, 5_2^+, 6_2^-) & (3_1^+, 5_2^+, 4_1^+, 6_2^-) & (3_1^+, 5_2^+, 6_2^-, 4_1^+) \\ (5_2^+, 3_1^+, 4_1^+, 6_2^-) & (5_2^+, 3_1^+, 6_2^-, 4_1^+) & (5_2^+, 6_2^-, 3_1^+, 4_1^+) \end{array}$$

represent the same order under OPP.

The full amplitude can be color decomposed to

$$\begin{aligned} \mathcal{A}(1^\Phi 2^{\Phi^\dagger}, n_1, n_2) &= \sum_{P(n_1), P(n_2)} (T_1^{a_1} T_1^{a_2} \cdots T_1^{a_{n_1}})_{ij} (T_2^{b_1} T_2^{b_2} \cdots T_2^{b_{n_2}})_{\bar{j}\bar{i}} \\ &\times A[1^\Phi 2^{\Phi^\dagger}, P(n_1), P(n_2)] \end{aligned}$$

- $\Phi V_2 \Phi^\dagger V_1$

$$A[1^\Phi 2^{\Phi^\dagger} 3_1^+ 4_2^-] = \frac{\langle 14 \rangle \langle 24 \rangle}{\langle 13 \rangle \langle 23 \rangle}$$

- $\Phi V_2 \Phi^\dagger V_1 V_1$

$$A[1^\Phi 2^{\Phi^\dagger} 3_1^+ 4_1^+ 5_2^-] = (-1) \frac{\langle 25 \rangle^2 \langle 15 \rangle^2}{\langle 23 \rangle \langle 34 \rangle \langle 41 \rangle \langle 25 \rangle \langle 51 \rangle}$$

- Compact formula for general case

$$A = \frac{\langle 2a \rangle^2 \langle 1a \rangle^2}{\underbrace{\langle 2\star \rangle \cdots \langle \star 1 \rangle}_{SU(N_1)} \underbrace{\langle 2* \rangle \cdots \langle *1 \rangle}_{SU(N_2)}}$$

Green: Particle with – helicity

Blue: Particle belongs to the first gauge group

Red: Particle belongs to the second gauge group

★: Order for gauge group 1

*****: Order for gauge group 2

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For example, if we want to compute $A[1^{\Phi} 2^{\Phi^{\dagger}} 5_1^{+} 3_1^{+} 4_1^{-} 7_2^{+} 6_2^{+} 8_2^{+}]$:

$$A = \frac{\langle 24 \rangle^2 \langle 14 \rangle^2}{\langle 25 \rangle \langle 53 \rangle \langle 34 \rangle \langle 41 \rangle \langle 27 \rangle \langle 76 \rangle \langle 68 \rangle \langle 81 \rangle}$$

If you use Feynman diagrams, it may take several days to accomplish the computation.

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- Review the on-shell method, including BCFW recursion relation, spinor-helicity formalism, color-ordering and etc.
- Review a (de)constructed gauge theory model, which is an effective field theory for 5 dimension gauge theory.
- Much of the scattering amplitudes in this model can be recursively computed by BCFW, and the color-ordered amplitudes can be well defined.

In the 1960s, Chew, Mandelstam and Weinberg proposed the **S-matrix Bootstrap**, seeking to determine scattering amplitudes from *Lorentz invariance*, *unitarity*, *analyticity*, *causality* and crossing symmetry, without reference to a Lagrangian.

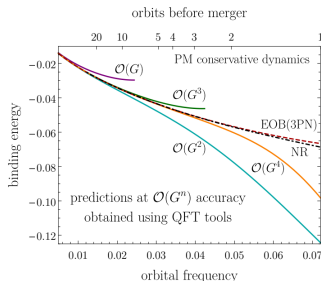
In the late 2000s, the **On-Shell Program**—developed by Nima Arkani-Hamed and collaborators—extended this philosophy to a practical computational framework, with tools like spinor-helicity, BCFW, generalized unitarity.

“The On-Shell Program is in some sense a modern reincarnation of the S-matrix Bootstrap, but now with tools powerful enough to actually compute.” — Nima Arkani-Hamed

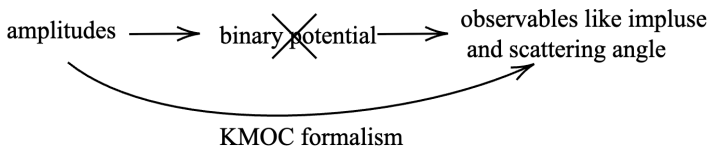
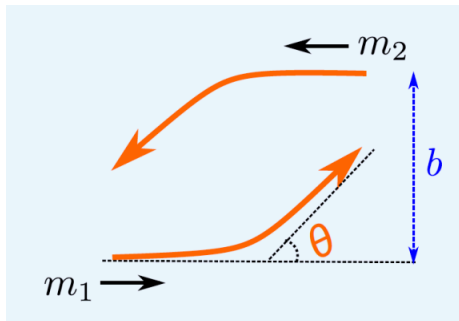
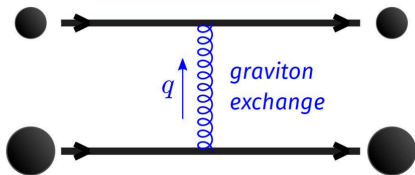
In recent years, **gravitational amplitudes** have attracted significant attention, particularly in applications to **binary systems** relevant for gravitational wave physics (*arXiv:2101.07254*, *Z. Bern et al.*).

Thanks for your attention!

Amplitude Methods for Inspiral Stage



- In relativistic QFT, perturbative expansion in g is analogous to post-Minkowskian expansion in GR (expansion in G).
- Effective field theory: widely separated black holes $\Rightarrow$ approximated by point particles.
 - Schwarzschild BH: modeled by massive scalar particles coupled to gravity.
 - Kerr BH: spin effects modeled by massive spin-1/2, spin-1, or higher-spin particles (can be handled by spinor-helicity)
- Advanced loop amplitude techniques from particle physics $\Rightarrow$ calculations beyond best classical results.



For merger process, there is an article discussing about it from K. Aoki, A. Cristofoli and Y. t. Huang, "On-shell approach to black hole mergers,"[[arXiv:2410.13632 \[hep-th\]](https://arxiv.org/abs/2410.13632)].

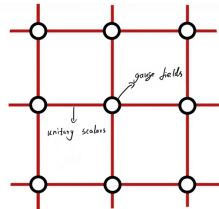
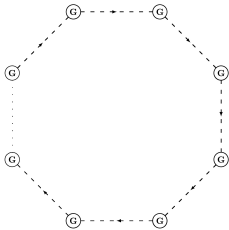
- nV_1 or nV_2

This part is completely the same as the pure gluon amplitude, so we can directly borrow the existing results.

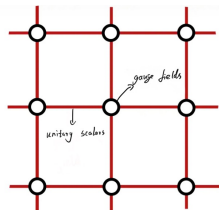
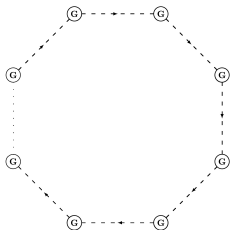
$$\text{Parke - Talyor Formula : } A[\cdots i^- \cdots j^- \cdots] = \frac{\langle ij \rangle^4}{\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle}$$

Notice that this formula only applies to MHV amplitudes, although the NMHV can be completely solved.

It has been proposed that this model actually discretized a **five-dimension gauge theory** with gauge group $SU(m)$, where only the fifth dimension are latticed. So it is an effective theory for 5d gauge theory.



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After higgsing the scalar field, we can obtain a spectrum

$$M_k^2 = 4g^2 f_s^2 \sin^2 \left(\frac{\pi k}{N} \right)$$

This is precisely the **Kaluza-Klein** spectrum under S^1 compactification.

Helicity

Helicity is defined as the projection of a particle's spin vector $\vec{S}$ onto the direction of its momentum $\vec{p}$:

$$h = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}$$

Spinor-Helicity Formalism

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S-matrix is a function of momentum p_i and helicity h_i

$$\mathcal{M}(p_i, h_i)$$

How can we catch the information of helicity?

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How can we catch the information of helicity?

Massless Case:

- Momenta in spinor form:

$$p_\mu \sigma^\mu = p_{\alpha\dot{\alpha}} = p_\alpha \tilde{p}_{\dot{\alpha}} = |p\rangle[p|$$

Brief explanation: We choose two momentum to be shifted oppositely

$$p_i \rightarrow \hat{p}_i(z) \equiv p_i - zk, \quad p_j \rightarrow \hat{p}_j(z) \equiv p_j + zk$$

satisfying

$$k^2 = 0, \quad p_i \cdot k = 0, \quad p_j \cdot k = 0$$

We consider amplitude A_n in terms of shifted momentum $\hat{p}_i^\mu$ instead of original real momentum.

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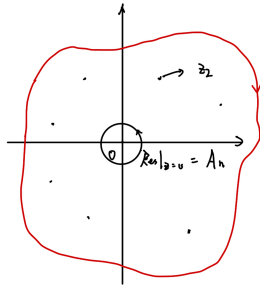
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$$A_n \longrightarrow \hat{A}_n(z)$$

If we consider the meromorphic function $\frac{\hat{A}_n(z)}{z}$ in the complex plane. From Cauchy Theorem, we can obtain

$$A_n = - \sum_{z_I} \text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n,$$

where B_n is the residue of the pole at $z = \infty$, called boundary term.



★ The most important point here is that

$$\text{Res}|_{z=0} \frac{\hat{A}_n(z)}{z} = \hat{A}_n(0) = A_n$$

and

$$\text{Res}|_{z=z_I} = \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I)$$

Large z behavior

In the BCFW formula, the boundary term B_n affects a lot

$$A_n = - \sum_{z_I} \text{Res}|_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n,$$

In most applications, one assumes or much better, proves $B_n = 0$. This is often justified by declaring a stronger condition

$$\hat{A}_n(z) \rightarrow 0 \quad \text{for} \quad z \rightarrow \infty$$

Here I show the large z behavior for gluon scattering

$[i \setminus j]$	+	-
+	$1/z$	z^3
-	$1/z$	$1/z$

On-shell 3-point can be completely determined

For the complex momentum, we have

$$|1\rangle \propto |2\rangle \propto |3\rangle \quad \text{or} \quad |1] \propto |2] \propto |3]$$

$A_3^{h_1 h_2 h_3} = c \langle 12 \rangle^{h_3 - h_1 - h_2} \langle 31 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3}$	$h_1 + h_2 + h_3 < 0$
$A_3^{h_1 h_2 h_3} = c' [12]^{h_1 + h_2 - h_3} [23]^{h_2 + h_3 - h_1} [31]^{h_3 + h_1 - h_2}$	$h_1 + h_2 + h_3 > 0$

★ **All massless on-shell 3-point amplitudes are completely determined by little group scaling!**

Example: 3-gluon amplitude

$$A_3(g_1^-, g_2^-, g_3^+) = g \frac{\langle 12 \rangle^3}{\langle 23 \rangle \langle 31 \rangle}$$

Scattering Amplitudes from BCFW

For simplicity, we start from the two-site gauge theory with gauge fields V_1, V_2 and scalar fields $\Phi, \Phi^\dagger$.

$$\mathcal{L} = -\frac{1}{2}\text{Tr}(F_1)^2 - \frac{1}{2}\text{Tr}(F_2)^2 + \text{Tr}[(D_\mu\Phi)^\dagger(D^\mu\Phi)],$$

We only focus on the following amplitudes:

$$nV_1, \quad nV_2, \quad \Phi^\dagger nV_1 \Phi, \quad \Phi nV_2 \Phi^\dagger, \quad \Phi^\dagger \Phi \Phi^\dagger \Phi$$

here n can be any positive integer.